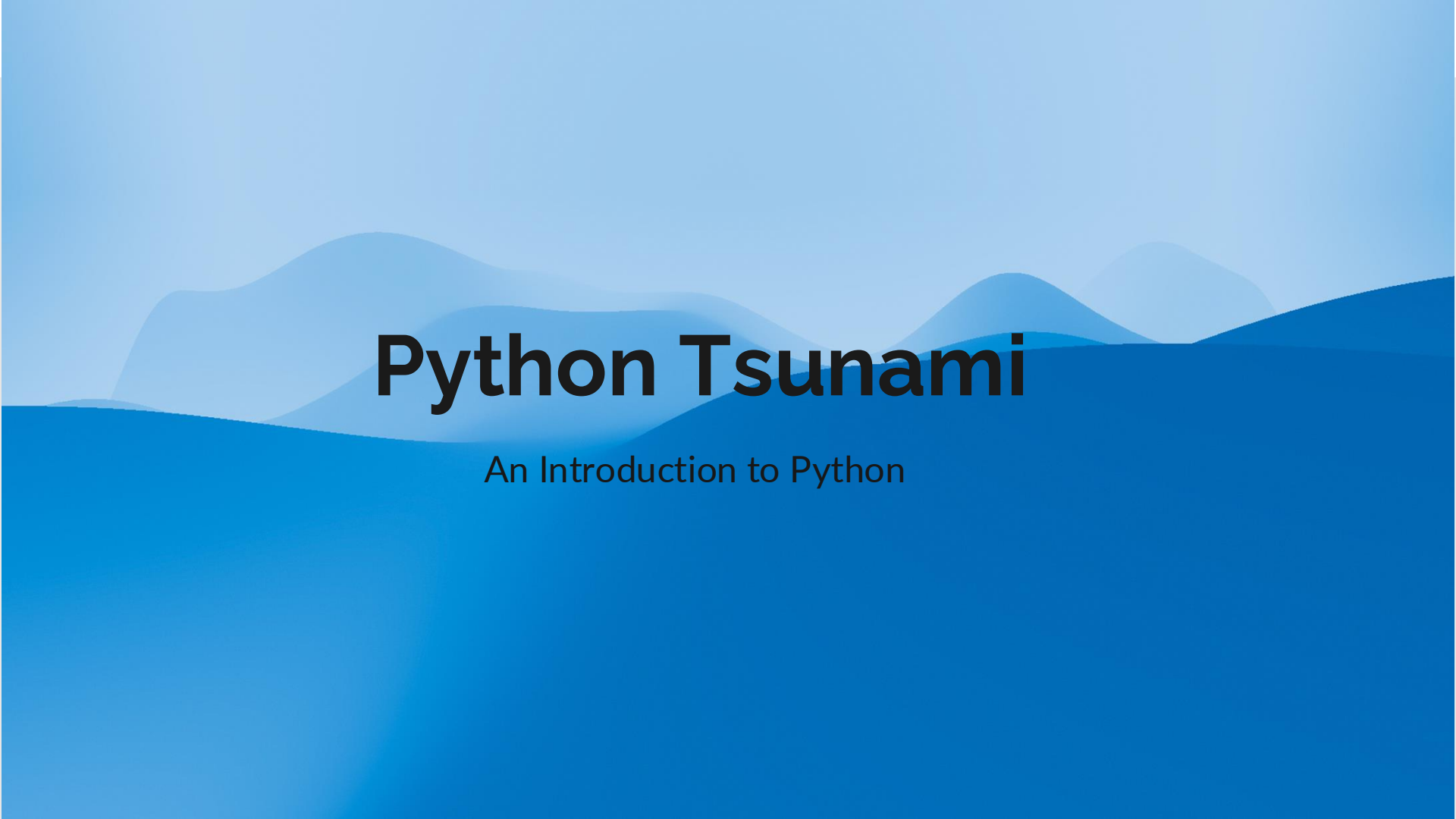


The background features a stylized landscape with blue waves in the foreground and rolling hills in the background. The text is centered over this graphic.

Python Tsunami

An Introduction to Python



Who are we?

Center for Health Data Science



Center for Health Data Science (HeaDS)

The mission of HeaDS is to strengthen health data science within the Faculty:

- Visible hub for Health Data Science
- Research within the field of HDS
- SUND Data Lab



<https://heads.ku.dk>



COURSES & WORKSHOPS

Data Science skills, Tools
and HDS Topics

COMMISSIONED RESEARCH

Commissioned Data Science Analysis
Commissioned Supervision



CONSULTATIONS

Need guidance? Drop by for a
consultation on your research
project

MATCHMAKING

Conference, seminars and
networking events - Join us!

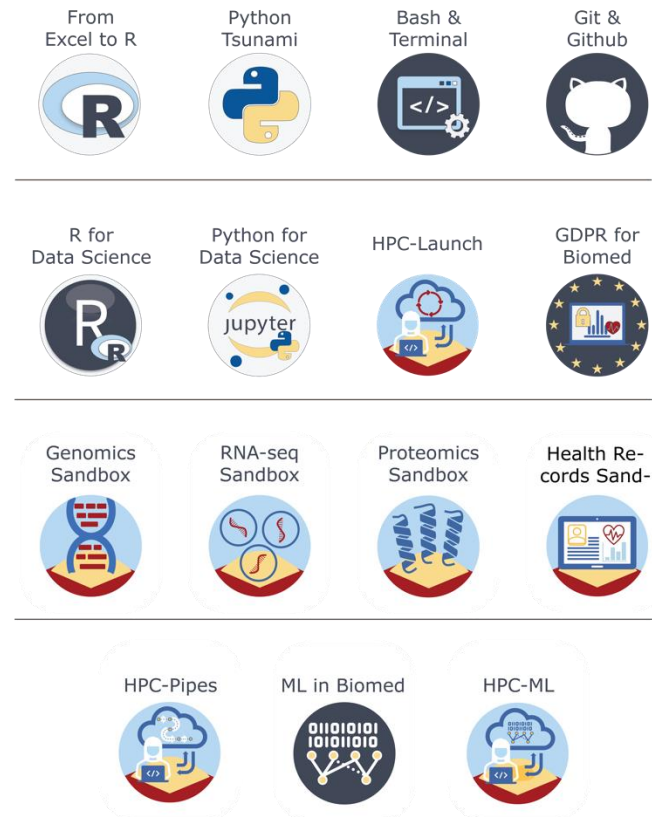


HeaDS Courses

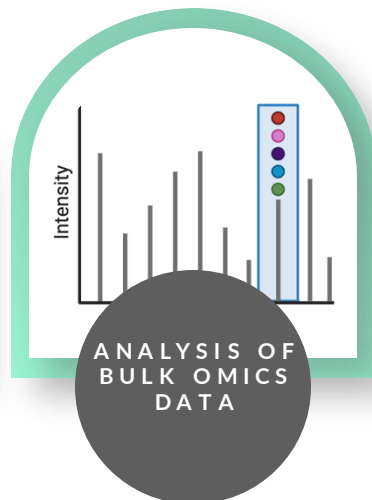
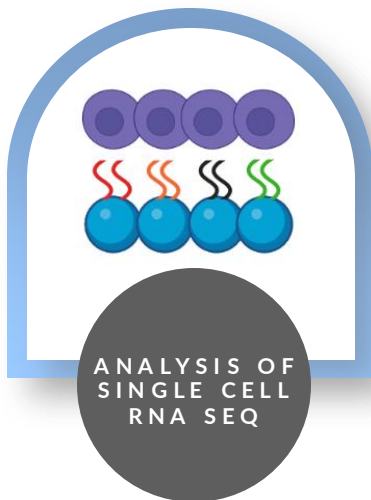


Skill Sets at Different Levels:

- Programming (R, Python)
- Omics Data Analysis
- Technical skills (Bash, Git)
- High Performance Compute Tools
- Machine Learning for HDS



Commissioned Research



- Machine Learning
- Biometric Data
- Metagenomics
- Epigenetics
- Biological Networks
- Seq. Motif Discovery
- Text mining
- *Biostatistical Models*

The Python Tsunami Team



Your teachers:

Lectures:

Thilde Terkelsen

Valentina Sora

Inigo Prada

Chloe Pittman



TAing:

Astrid Filt Beyer

Yan Li

Suze Rooster



Original Content:

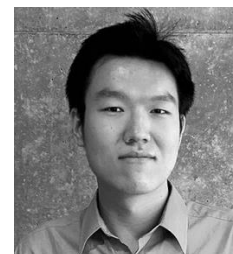
Henrike Zschach (Research IT)

Alberto Delgado (DTU, Biosustain),

Henry Webel (DTU, Biosustain),

Annelaura Nielsen (University Hospital),

Rita Colaço (PRI)





About this course

Day 1		Day 2		Day 3	
08:30-08:45	Morning Coffee	08:30-08:45	Morning Coffee	08:30-08:45	Morning Coffee
08:45-09:15	Intro to Python & Colab	08:45-10:15	Pandas: Series, Dataframes, Indexing	08:45-09:50	Recap quiz + Q&A
09:15-09:50	Variables & Data Types				
09:50-10:00	Break	10:15-10:30	Break	09:50-10:00	Break
10:00-10:50	Iterables I: Lists	10:30-12:00	Pandas: Data selection & Summary	10:00-12:00	Visualization
10:50-11:00	Break				
11:00-12:00	Iterables II: Sets, Dicts & Tuples				
12:00-13:00	Lunch	12:00-13:00	Lunch	12:00-13:00	Lunch
13:00-14:15	Booleans, Operators & Conditions	13:00-14:00	Pandas: Modifying Data	13:00-13:30	Local Installations
14:15-14:30	Break	14:00-14:15	Break	13:30-16:00	Dataset Exercise
14:30-16:00	Loops	14:15-15:30	Pandas: Groupby, Sort & Impute		
		15:30-16:00	Pandas Q&A		
16:00	See you tomorrow!	16:00	See you tomorrow!	16:00	Course Finished!



Python

Why Python?

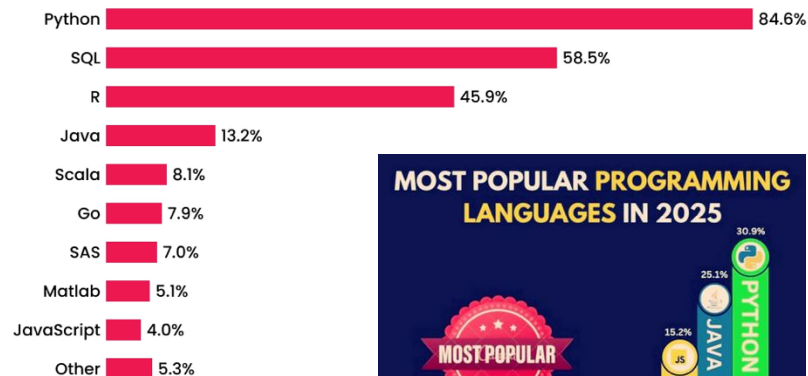


Python is a great programming language for both beginners and advanced programmers:

- Easy to grasp, close to natural language
- Many learning resources available
- Large community
- Libraries
- Simple or very advanced analysis

Required Programming Languages

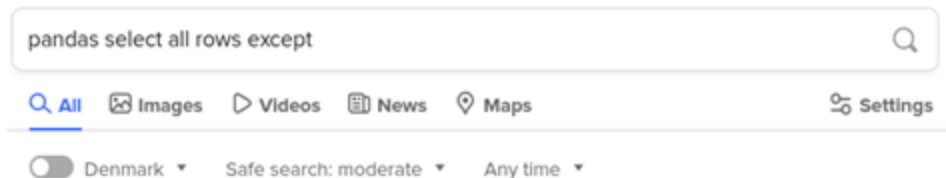
Data Scientists 2025



Where to get help



Online communities such as **stackoverflow** are an important tool in programming.

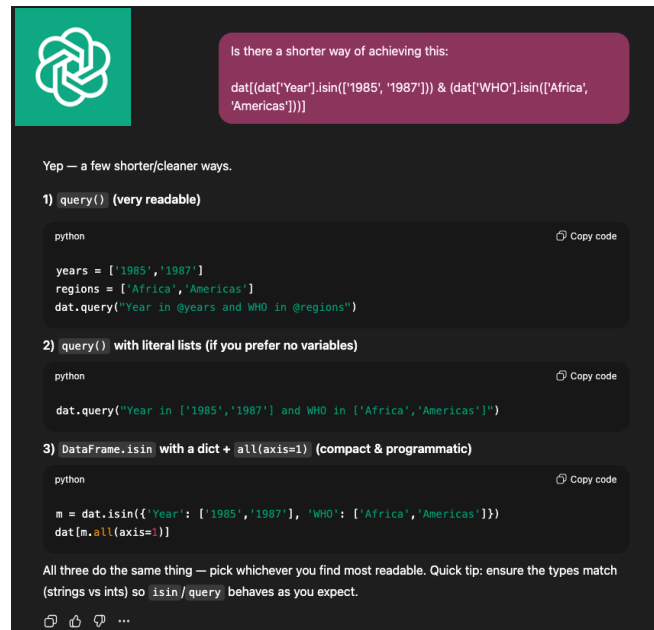


[https://stackoverflow.com > questions > 28256761 > select-pandas-rows-by-excluding-index-num...](https://stackoverflow.com/questions/28256761/select-pandas-rows-by-excluding-index-num...)

select pandas rows by excluding index number - Stack Overflow

I'm looking to slice a **Pandas** dataframe by using index numbers. I have a list/core index with the index numbers that I do NOT need, shown below **pandas**. Stack Overflow ... Use a list of values to **select rows** from a **Pandas** dataframe. 2015. Delete a column from a **Pandas** DataFrame. 1290. How to drop **rows** of **Pandas** DataFrame whose value in a certain ...

Or ask **ChatGPT**. But remember: “With great power comes great responsibility!”

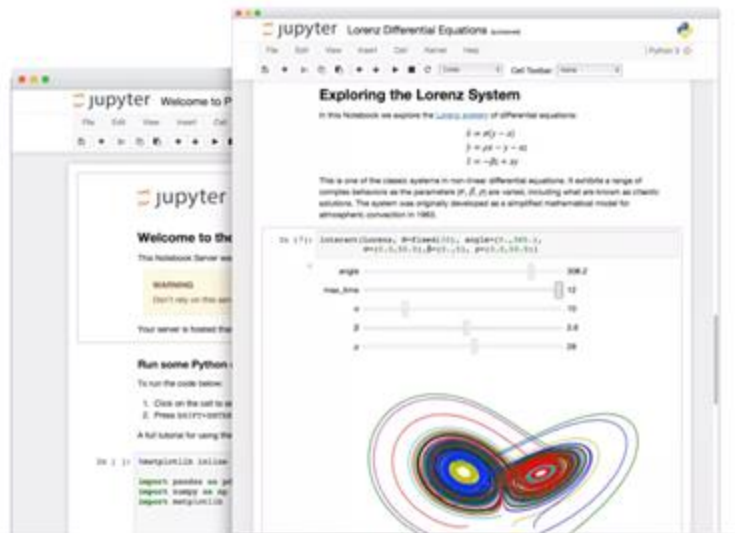


Python Environments

Jupyter notebook



Jupyter Notebook is an **open-source application** to create and share documents that contain code, visualizations and text (markdown).



- Browser-based development environment for creating, running and sharing Python code
- Combine code with text and output
- Runs from your **local installation**. I.e., you need Python and the libraries you want to use installed on your computer



Google Colab is a Jupyter Notebook hosted on Google's servers, not your own machine. It still runs in your browser.

- Tool to write, execute and share Python code through the browser
- Requires no setup to use and provides free access to computing resources on Google's servers including GPUs
- Connected to a Google account and data and notebooks can be accessed through Google Drive.

We'll use Colab during the course.

Python Libraries

Python has many libraries, also called packages, that other programmers have developed. Find and **use** them!

Well-maintained libraries generally are:

- Tested
- Optimized
- Documented

There is no need to reinvent the wheel. During this course we will use:

- Pandas (all the data analysis!)
- Math (basic math)
- Plotly express (visualization)

Running Python from a local installation requires you to have libraries **downloaded & installed** before you can use them.

On Google Colab you can generally just import, they are already installed.

- Import the math library:

```
import math
```

- Now I can use functions from that library, i.e. calculating the logarithm or square root:

```
math.log(3)
```

```
math.sqrt(4)
```

Course Material



Center-for-Health-Data-Science / PythonTsunami Public

forked from [pythontsunami/teaching](#)

[Code](#) [Issues](#) 10 [Pull requests](#) [Discussions](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

spring2022 16 branches 0 tags

[Go to file](#) [Add file](#) [Code](#)

This branch is 111 commits ahead of [pythontsunami/teaching:heads](#). [Contribute](#) [Fetch upstream](#)

hezscha Add files via upload 678c67a 2 days ago 376 commits

Conditionals	Add files via upload	6 days ago
Exercise	Add files via upload	2 days ago
Functions	Add files via upload	2 days ago
Introduction_and_tools	Add files via upload	7 days ago
Iterables	Add files via upload	2 days ago
Loops	Add files via upload	2 days ago
Pandas	Minor changes to pandas examples	6 days ago
Recap	Add files via upload	9 days ago
Variables_data_types	Add files via upload	6 days ago

You can find the course material here:

<https://github.com/Center-for-Health-Data-Science/PythonTsunami>

Course Material



spring2022 PythonTsunami / Variables_data_types /

This branch is 111 commits ahead of pythontsunami/teaching/heads.

hezscha Add files via upload ab1c127 6 days ago History

..

README.md	Minor fixes	7 days ago
variables.ipynb	Update Colab link within the notebook	6 days ago
variables_solutions.ipynb	Add files via upload	6 days ago

README.md

HeaDS

Python Tsunami

notebook	content
variables.ipynb  Open in Colab	Variables and data types



Course Material



Remember to **save a copy** to your own google drive so you can save your notes and exercises!



Short Introduction



Take the next 5 mins to introduce yourself at your table:

- Name
- Position/Unit
- What you do (very briefly!)
- Why would you like to learn Python?